

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Enterprise Java

M2:Practical – IX

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Class: TYIT	Batch: 1
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9. Implement the following JPA applications.

a. Develop a simple Inventory Application Using JPA.

CheckInventory.jsp:

```
<%@page import="java.util.Iterator"%>
<%@page import="java.util.List"%>
<%@page import="javax.persistence.Persistence"%>
<%@page import="javax.persistence.EntityManager"%>
<%@page
import="javax.persistence.EntityManagerFactory"%>
<%@page
import="javax.persistence.EntityTransaction"%>
<%@page import="myAPP.Inventory"%>
<%@page contentType="text/html" pageEncoding="UTF-
8"%>
<%
EntityManagerFactory eMF =

Persistence.createEntityManagerFactory("INVJPAPU");
EntityManager eM = eMF.createEntityManager();
EntityTransaction eT = null;
List<Inventory> inventoryList;
String submit = request.getParameter("AddToInventory");
if (submit != null && submit.equals("AddToInventory")) {
    try {
        String itemname =
            request.getParameter("itemname");
        String description =
            request.getParameter("description");
        String quantity1 =
            request.getParameter("quantity");
        int quantity =
            Integer.parseInt(quantity1);
        String addedDate =
            new java.util.Date().toString();
        Inventory inv = new Inventory();
        inv.setItemName(itemname);
        inv.setDescription(description);
        inv.setQuantity(quantity);
        inv.setAddedDate(addedDate);
        eT = eM.getTransaction();
        eT.begin();
        eM.persist(inv);
        eT.commit();
    } catch (RuntimeException e) {
        if (eT != null && eT.isActive()) {
            eT.rollback();
        }
        throw e;
    }
}
```

```
}
inventoryList =
    eM.createQuery("SELECT i FROM Inventory i",
        Inventory.class)
        .getResultList();
eM.close();
%>
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <title>Inventory</title>
</head>
<body>
<table border="1">
<tr>
    <td colspan="5" align="center">
        <b>
            Click
            <a href="index.jsp">here</a>
            to add items in the Inventory.
        </b>
    </td>
</tr>
<tr>
    <th>Item ID</th>
    <th>Item Name</th>
    <th>Description</th>
    <th>Quantity</th>
    <th>Added Date & Time</th>
</tr>
<%
Iterator<Inventory> iterator =
    inventoryList.iterator();
while (iterator.hasNext()) {
    Inventory obj =
        iterator.next();
%>
<tr>
    <td><%=obj.getItemID()%></td>
    <td><%=obj.getItemName()%></td>
    <td><%=obj.getDescription()%></td>
    <td><%=obj.getQuantity()%></td>
    <td><%=obj.getAddedDate()%></td>
</tr>
<%
}
%>
</table>
</body>
</html>
```

Index.jsp:

```

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <title>Inventory</title>
</head>
<body>
<form action="checkInventory.jsp" method="post">
<table>
<tr>
    <td colspan="2" align="center">
        <b>Add the item in Inventory</b>
    </td>
</tr>
<tr>
    <td>Item Name:</td>
    <td>
        <input name="itemname" maxlength="25" size="50">
    </td>
</tr>
<tr>
    <td>Description:</td>
    <td>
        <textarea rows="5" cols="36"
            name="description"></textarea>
    </td>
</tr>
<tr>
    <td>Quantity:</td>
    <td>
        <input name="quantity" maxlength="4" size="5">
    </td>
</tr>
<tr>
    <td colspan="2">
        <input type="submit"
            name="AddToInventory"
            value="AddToInventory">
    </td>
</tr>
</table>
</form>
</body>
</html>

```

Inventory.jsp:

```

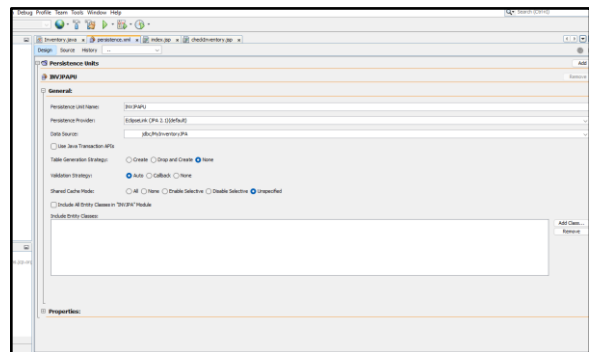
package myAPP;
import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.GeneratedValue;
import javax.persistence.GenerationType;
import javax.persistence.Id;
import javax.persistence.Table;
@Entity
@Table(name = "inventory")
public class Inventory {
    @Id
    @GeneratedValue(strategy =
        GenerationType.IDENTITY)
    @Column(name = "itemid", unique = true, updatable =
        false)

```

```

private Integer itemid;
@Column(name = "itemName")
private String itemName;
@Column(name = "description")
private String description;
@Column(name = "quantity")
private Integer quantity;
@Column(name = "addedDate")
private String addedDate;
public Inventory() {
}
public Integer getItemID() {
    return itemid;
}
public void setItemID(Integer itemid) {
    this.itemid = itemid;
}
public String getItemName() {
    return itemName;
}
public void setItemName(String itemName) {
    this.itemName = itemName;
}
public String getDescription() {
    return description;
}
public void setDescription(String description) {
    this.description = description;
}
public Integer getQuantity() {
    return quantity;
}
public void setQuantity(Integer quantity) {
    this.quantity = quantity;
}
public String getAddedDate() {
    return addedDate;
}
public void setAddedDate(String addedDate) {
    this.addedDate = addedDate;
}
}

```



New Connection Wizard

Customize Connection

Driver Name: MySQL (Connector/J driver)

Host: localhost Port: 3306

Database: inventory

User Name: root

Password: ****

☐ Remember password

Connection Properties Test Connection

JDBC URL: jdbc:mysql://localhost:3306/inventory?zeroDateTimeBehavior=convertToNull

< Back Next > Finish Cancel Help

New Persistence Unit

Steps

1. Choose File Type
2. Provider and Database

Provider and Database

Persistence Unit Name: INVJPA

Specify the persistence provider and database for entity classes.

Persistence Provider: EclipseLink (PA 2.1)(default)

Data Source: New Data Source...

Create Data Source

JNDI Name: jdbc/MyInventoryJPA

Database Connection: jdbc:mysql://localhost:3306/inventory?zeroDateTimeBehavi...

OK Cancel Help

< Back Next > Finish Cancel Help

localhost:8080/INVJPA/

Item Name: laptop

Description: Dell laptop

Quantity: 10

AddToInventory

Click [here](#) to add items in the Inventory.

Item ID	Item Name	Description	Quantity	Added Date & Time
1	laptop	Dell laptop	10	Fri Sep 11 13:46:22 IST 2026
2	laptop	Dell laptop	10	Fri Sep 11 13:47:06 IST 2026

```
mysql> USE Inventory;
Database changed
mysql> CREATE TABLE inventory (
->     itemid INT PRIMARY KEY AUTO_INCREMENT,
->     itemName VARCHAR(50),
->     description VARCHAR(200),
->     quantity INT,
->     addedDate VARCHAR(40)
-> );
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_inventory |
+-----+
| inventory            |
+-----+
1 row in set (0.00 sec)

mysql>
```